EX. NO: 12

NMAP TO DISCOVER LIVE HOST USING ARP SCAN, ICMP SCAN AND TCP/UDP PING SCAN

AIM

To learn and implement Nmap host discovery techniques using ARP, ICMP, TCP and UDP ping scans, and compare their effectiveness.

PROCEDURE

1. Find the target subnet.
2. Run an ARP scan: nmap -sn -PR 10.10.5.0/24
3. Run an ICMP scan: nmap -sn -PE 10.10.5.0/24
4. Run a TCP SYN scan: nmap -sn -PS22,80,443 10.10.5.0/24
5. Run a UDP scan: nmap -sn -PU53 10.10.5.0/24
6. Compare the number of hosts found by each method.

OUTPUT

```
$ nmap -sn -PR 10.10.5.0/24
Nmap scan report for 10.10.5.1
Host is up (0.00071s latency).
Nmap scan report for 10.10.5.12
Host is up (0.00098s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 2.14 seconds

$ nmap -sn -PU53 10.10.5.0/24
Nmap scan report for 10.10.5.1
Host is up (0.0022s latency).
Nmap done: 256 IP addresses (1 host up) scanned in 4.88 seconds
```

RESULT

Live hosts were discovered using ARP, ICMP, TCP and UDP ping scans, and ARP scan was found to be the most effective on the local subnet.